

Sarthak Kanoi

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Summary

Software Engineer | Codeforces Candidate Master

Software Engineering undergraduate with a strong foundation in advanced algorithms and low-latency architecture. Proven track record in competitive programming, ranking in **top 1.5% in Codeforces Round 1091**. Experienced in building concurrent systems, designing optimized APIs, and applying robust data structures to solve complex problem-solving workflows.

Competitive Programming & Achievements

- **Codeforces Candidate Master** (Max Rating: 1919) [[Codeforces Profile](#)]
- **LeetCode** (Max Rating: 1834, Top 6.6% globally) [[LeetCode Profile](#)]
- Secured **Global Rank 235** out of **14,802** participants (**Top 1.58%**) in Codeforces Round 1091 (Div 2) [[Link](#)]
- Secured **Global Rank 522** out of **37,342** participants (**Top 1.39%**) in LeetCode Biweekly Contest 178 [[Link](#)]
- **Advanced Problem Solving:** Solved **1000+** **Data Structures and Algorithms** problems across platforms including **LeetCode, Codeforces, CodeChef and GeeksforGeeks**, covering advanced patterns like Graphs, DP, and Greedy Algorithms

Education

Raj Kumar Goel Institute of Technology, Ghaziabad

B.Tech in Computer Science and Engineering (IoT)

R.P.M Academy, Gorakhpur

Senior secondary, PCM

Aug 2023 - July 2027

CGPA : 8.2/10

July 2021 - May 2023

Percentage : 87%

Projects

FAANG-Forge – Interview Preparation Platform | [[Live](#)] | [[GitHub](#)]

Jan 2026 – Mar 2026

React, Node.js, Express, MongoDB, JWT, Vercel, Render

- **Engineered a structured preparation platform** utilizing a custom-designed spaced-repetition algorithm to calculate dynamic problem-revisit intervals, directly improving long-term retention metrics for **25+ registered users**.
- **Designed and integrated a robust ETL-style pipeline** with an external GraphQL API to dynamically fetch, parse, and categorize **7,715+ company-tagged problems** across 463 companies.
- **Architected a RESTful API backend** secured via JWT authentication and optimized MongoDB schemas to handle high-frequency user progress tracking and relational query filtering with low latency.

MockMate AI – AI Voice Interview Coach | [[Live](#)] | [[GitHub](#)]

Oct 2025 – Dec 2025

Next.js 14, TypeScript, VAPI (Voice API), MongoDB, NextAuth.js, Vercel

- **Developed a real-time, bi-directional audio streaming platform** using WebSockets, optimizing data packet transmission to achieve low-latency voice communications for mock interviews.
- **Implemented secure user authentication and stateful session persistence** utilizing NextAuth.js and MongoDB Atlas, safely caching user interview logs and structured AI feedback strings.
- **Built a dynamic, condition-based routing system** that evaluates technical focus areas and user-selected seniority tiers to programmatically serve tailored, algorithmic interview prompts.

DevSync – Real-Time Collaborative IDE | [[Live](#)] | [[GitHub](#)]

July 2025 – Sept 2025

Next.js, TypeScript, Socket.IO, Piston API, Monaco Editor, Railway

- **Architected a multi-room collaborative code editor** utilizing Socket.io, implementing event-driven state synchronization to support seamless, simultaneous cursor tracking and text updates for multiple concurrent users.
- **Integrated the isolated Piston API** into the backend execution layer, securely sandboxing and processing code compilation requests for **40+ programming languages** directly via the browser interface.
- **Designed an efficient room-based communication model** with unique shareable URLs, managing transient WebSocket states to reliably synchronize a shared vector canvas whiteboard alongside the text buffer.

Technical Skills

Languages: Java, C++, JavaScript (ES6+)

Core Concepts: Data Structures & Algorithms, System Design, OOPs, DBMS, OS, CN

Backend & Databases: Node.js, Express, RESTful API Design, JWT, MongoDB Atlas, Authentication (JWT/OAuth)

Frontend: React, Next.js, Tailwind CSS

Tools & Infrastructure: Git/GitHub, VS Code, WebSockets, Railway, Vercel, Render